

Suyash Bandi

(+91) 8717994401 | suyashbandi16@gmail.com | linkedin.com/in/suyashbandi | github.com/Suyash-16

TECHNICAL SKILLS

C++, JavaScript, HTML, CSS, React.js, Tailwind, Node.js, Express.js, WebSockets, RESTful APIs, MongoDB, MySQL, Git, GitHub, FFmpeg, HLS/DASH Streaming

PROJECTS

FlowCast – Video Streaming Platform | Node.js, Express.js, React.js, FFmpeg, REST APIs

- Built a full-stack video streaming platform enabling video upload, transcoding, and adaptive playback using FFmpeg, Node.js, Express.js, and React.js.
- Implemented multi-resolution streaming (240p–1080p) for seamless playback across devices.
- Designed a custom video player with thumbnails, playback controls, and video metadata display for improved user experience.
- Developed REST APIs for efficient video management and ensured optimized performance during streaming and playback.

CollaBoard – Real-Time Collaborative Whiteboard | React.js, Node.js, Express.js, Socket.IO

- Developed a real-time collaborative whiteboard enabling multiple users to draw simultaneously using stroke-based synchronization.
- Engineered efficient WebSocket communication via Socket.IO for smooth, low-latency multi-user updates.
- Built intuitive drawing tools (pencil, line, rectangle), color picker, and undo/redo functionality using the HTML5 Canvas API.
- Implemented room-based collaboration for isolated sessions with persistent socket connections and event-driven updates.

AES Encryption Algorithm Implementation — C++

- Implemented the AES-128 symmetric encryption algorithm to securely encrypt and decrypt files.
- Developed modular C++ code to handle key generation, substitution, permutation, and round transformations.
- Tested the program with multiple file types to ensure data integrity and correctness of encryption/decryption.
- Measured execution time: 1368 microseconds (1 millisecond) per encryption/decryption operation.
- Documented the methodology, explaining each AES step and its role in secure data communication.

ACHIEVEMENTS & CO-CURRICULAR ACTIVITIES

Adobe India Hackathon – Round 1: Performed structured analysis of unstructured documents using heuristics and machine learning.

NPTEL Certification – Introduction to Algorithms and Analysis (IIT Kharagpur), Jul–Oct 2024, achieving a consolidated score of 78%.

Event Team Chief, VITronix Club, VIT Bhopal (2024-25): Organized and managed three major technical events with 300+ participants, ensuring successful execution and high engagement.

Solved over 200 problems on LeetCode to strengthen data structures and algorithms.

EDUCATION

Vellore Institute of Technology Bhopal

Bachelor of Technology in Computer Science and Engineering

Bhopal, MP

June 2022 – Present

St Arnolds Vijaynagar Indore

Senior Secondary (10th and 12th)

Indore, MP

May 2020/July 2022